



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	2 OCTOBER 2023
GRADE	11	MARKS	40
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	40 MINUTES
CLASS			

MARKING GUIDELINES

Question 1:

- 1. E ✓
- H ✓
- B ✓
- G ✓
- C ✓
- A ✓
- J ✓
- F ✓
- I ✓
- D ✓

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Question 2

- 2.1. Type of bacteria ✓ (1)
- 2.2. All bacteria ✓ will be killed by penicillin. ✓ (any **statement** regarding the independent and dependent variable) (2)
- 2.3. Bacteria C ✓ (1)
- 2.4. There was no change in the number of bacteria✓ after the penicillin was added and no clear zone formed✓. (2)
- 2.5. A few bacteria develop random mutations✓ which cause them to be resistant to an antibiotic. These resistant bacteria will survive and multiply✓ producing a fully antibiotic resistant population. ✓ (3)
- 2.6. The full course of antibiotics needs to be taken/ completed
Numerous courses/ ongoing antibiotics should be avoided ✓ (any 1)
- 2.7.
 - A. To compare results ✓ and see what the natural result would look like ✓ (2)
 - B. Penicillin ✓ (1)

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Question 3

3.1.

3.1.1. Protists/ protista ✓ (1)

3.1.2.

a. 40,2 °C ✓ (accept 40-40,5) (1)

b. 3 ✓ (1)

3.1.3. Malaria symptoms only appear after 10 days ✓ (1)

3.1.4. Day 17 ✓ (1)

3.1.5. Less oxygen will be delivered to the cells✓ and therefore a person will have less energy✓ / lowered respiration due to less oxygen
(1 mark if they just say "fatigue") (2)

3.1.6. Take preventative medication before entering a malaria area (prescribed malaria medication)
Wear long clothing in malaria areas
Use repellent sprays and lotions
Sleep under a mosquito net
Spray mosquito nets with repellent sprays
Close all windows and doors from dusk
Stay indoors from dusk until dawn
Any reasonable, logical answer ✓✓ (any 2)

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3.2.

3.2.1. They start their immune response. (1)

3.2.2. A person will be classified as having AIDs when the CD4 cells have been reduced ✓ and can no longer fight the HIV virus and the virus keeps multiplying ✓
When the CD4 count is lower than 200 cells/ mL of blood
Opportunistic diseases have taken a toll on the body (any 2)

3.2.3. Because the HI virus attacks CD4 cells it stops the body from being able to start an immune response. ✓ Not being able to start an immune response the body becomes susceptible to any other type of infection ✓ and thus it won't be able to fight off the infection and thus the person will die. ✓ (3)

3.2.4. Bacteria can grow easily in favourable conditions ✓ whereas a virus needs a living host to grow in ✓ and they destroy the host so thus the virus will not reproduce more if there isn't more host tissue ✓ (any 2)

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TOTAL: 40 MARKS